

Domenico MANDAGLIO

Assistant Professor (RTDa) at University of Calabria

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RESEARCH INTERESTS

My research interests lie primarily in the broad areas of *artificial intelligence* and *(algorithmic) data science*, with the main goal of tackling (novel) problems and designing effective yet efficient algorithms that are useful to gain knowledge from data. Due to the fact that much of today's data can be represented as graphs, my emphasis has primarily been on formalizing data-driven challenges as graph problems and developing scalable methods to analyze complex relational structures. More specifically, my recent research interests include *graph mining and learning*, *explainable AI*, and *natural language processing*—the latter mainly in settings where graphs come with rich textual descriptions (e.g., node/edge metadata or associated documents) that can be leveraged to enhance graph representations and reasoning.

RESEARCH EXPERIENCE

Assistant Professor (RTDa)

University of Calabria, DIMES department

June 2023 - present
Rende, Italy

- Scientific supervisor: prof. Sergio Greco
- Area of research: polarization detection, graph representation learning, eXplainable AI

Postdoctoral Researcher

University of Calabria, DIMES department

May 2021 - June 2023
Rende, Italy

- Scientific supervisor: prof. Andrea Tagarelli
- Area of research: correlation clustering, combinatorial bandits, active learning

Research & Teaching Assistant

University of Calabria, DIMES department

October 2017 - May 2021
Rende, Italy

- Scientific supervisors: prof. Sergio Greco and prof. Andrea Tagarelli
- Area of research: community detection, clustering uncertain graphs, preference learning

Research Fellow

University of Calabria, DIMES department

April 2018 - June 2018
Rende, Italy

- Scientific supervisor: prof. Filippo Furfaro
- Area of research: privacy-preserving synopses of multidimensional data

EDUCATION

Ph.D. in Information and Communication Technologies

University of Calabria

October 2017 - May 2021
Rende, Italy

- Thesis: "Mining and Learning Problems in Complex Graph Data"
- Supervisors: prof. Sergio Greco and prof. Andrea Tagarelli

M.Sc. in Computer Science & Engineering (with honors)

University of Calabria

October 2014 - September 2017
Rende, Italy

- Thesis: "Ensemble-based Community Detection in Multilayer Networks"
- Supervisor: prof. Andrea Tagarelli

B.Sc. in Computer Science & Engineering (with honors)

University of Calabria

October 2011 - September 2014
Rende, Italy

- Thesis: "Implementation of Outlier Detection Algorithms in Weka"
- Supervisor: prof. Fabio Fassetti

SCIENTIFIC COMMUNITY SERVICE

Organization

- **Proceedings Co-Chair**, 19th ACM International Conference on Web Search and Data Mining (WSDM '26)
- **General Co-Chair**, Workshop on Mining Data for financial applications (MIDAS '25, MIDAS '26), in conjunction with the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD '25, ECML-PKDD '26)

- › **PhD Forum and Posters Track Co-Chair**, *16th International Conference on Advances in Social Networks Analysis and Mining (ASONAM '24)*

Program Committee (Major Conferences)

- › *ACM The Web Conference (WWW)* (Graph Algorithms and Modeling for the Web track) [2024, 2025, 2026]
- › *Conference on Neural Information Processing Systems (NeurIPS)* [2024, 2025, 2026]
- › *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)* (Journal track) [2026]
- › *International Conference on Machine Learning (ICML)* [2025]
- › *AAAI Conference on Artificial Intelligence (AAAI)* [2023, 2024, 2025]
- › *International Joint Conference on Artificial Intelligence (IJCAI)* [2024]
- › *ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)* (Research track) [2023]
- › *ACM The Web Conference (WWW)* (Social Network Analysis and Graph Algorithms track) [2022, 2023]
- › *ACM International Conference on Web Search and Data Mining (WSDM)* [2023]
- › *European Conference on Artificial Intelligence (ECAI)* [2023, 2024]
- › *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)* (Research track) [2022, 2023, 2024]
- › *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)* (Applied Data Science track) [2021, 2022]

Program Committee (Other Conferences & Workshops)

- › *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)* - special session on *Learning and Reasoning on Knowledge and Heterogeneous Graphs* [2026]
- › *Italian Symposium on Advanced Database Systems (SEBD)* [2026]
- › *Workshop on Sustainable Artificial Intelligence for addressing Online Information Disorder (SAI4OID)*, in conjunction with the *International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT)* [2025]
- › *Workshop on Mining Data for financial applications (MIDAS)*, in conjunction with the *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)* [2021, 2022, 2023, 2024]
- › *French Regional Conference on Complex Systems (FRCCS)* [2024]
- › *Workshop on Advances in Argumentation in Artificial Intelligence (AI^3)*, in conjunction with the *International Conference of the Italian Association for Artificial Intelligence (AIxIA)* [2023]
- › *International Symposium on Methodologies for Intelligent Systems (ISMIS)* [2022]

Conference Reviewing (The list does not include reviews done as a Program Committee member)

- › *ACM SIGMOD International Conference on Management of Data (SIGMOD)* [2023]
- › *ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)* [2019, 2020, 2021, 2022]
- › *International Joint Conference on Artificial Intelligence (IJCAI)* [2022]
- › *ACM The Web Conference (WWW)* [2019, 2020]
- › *IEEE International Conference on Data Mining (ICDM)* [2018, 2020, 2021]
- › *SIAM International Conference on Data Mining (SDM)* [2018]
- › *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)* [2018, 2019]

Journal Reviewing

- › *IEEE Transactions on Knowledge and Data Engineering (TKDE)*
- › *Data Mining and Knowledge Discovery (DAMI)*
- › *Machine Learning (MACH)*
- › *Discrete Mathematics (DISCM)*
- › *IEEE Transactions on Network Science and Engineering (TNSE)*
- › *International Journal of Human-Computer Studies (IJHCS)*
- › *Social Network Analysis and Mining (SNAM)*
- › *Applied Network Science (APNS)*
- › *Journal of Intelligent Information Systems (JIIS)*

Scientific Boards

- › Member of the Editorial Board of Data Mining and Management as Review Editor for Frontiers in Big Data [2021-2025]

DEPARTMENTAL/UNIVERSITY SERVICE

- › **Member of the Orientation and Internships Committee for the Unified Study Program in Computer Science & Engineering**
University of Calabria, DIMES department November 2023 - present
- › **Member of Degree Examination Committee** ("sedute di laurea")
University of Calabria, DIMES/DESF departments
Sessions: 2026 (April 20), 2024 (April 16, July 24, September 18, December 10), 2025 (April 29, April 30, October 17, December 10)

TALKS & POSTERS

Presenting Author

- › *Semifactual Explanations for GNN-based Classification: Formal Foundations, Complexity and Computation*, International Joint Conference on Artificial Intelligence (**IJCAI '26**), Bremen, Germany, August 2026
- › *Polarized Communities meet Densest Subgraph: Efficient and Effective Polarization Detection in Signed Networks*, Singapore ACM SIGKDD Symposium (**SIGKDD Singapore '26**), Singapore, January 2026
- › *Discovering Balance-Aware Polarized Communities in Signed Networks with Graph Neural Networks*, Italian Symposium on Advanced Database Systems (**SEBD '25**), Ischia, Italy, June 2025
- › *Link Prediction on Multilayer Networks through Learning of Within-Layer and Across-Layer Node-Pair Structural Features and Node Embedding Similarity*, ACM The Web Conference (**WWW '24**), Singapore, May 2024
- › *A Combinatorial Multi-Armed Bandit Approach to Correlation Clustering*, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD '23**), Turin, Italy, September 2023
- › *Learning to Active Learn by Gradient Variation Based on Instance Importance*, International Conference on Pattern Recognition (**ICPR '22**), Montréal, Québec, August 2022
- › *Correlation Clustering: from Local to Global Constraints*, Italian Symposium on Advanced Database Systems (**SEBD '22**), Tirrenia (Pisa), Italy, June 2022
- › *Correlation Clustering with Global Weight Bounds*, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD '21**), Virtual Event, September 2021
- › *In and Out: Optimizing Overall Interaction in Probabilistic Graphs under Clustering Constraints*, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD '20**), Virtual Event, August 2020
- › *A Combinatorial Multi-Armed Bandit Based Method for Dynamic Consensus Community Detection in Temporal Networks*, International Conference on Discovery Science (**DS '19**), Split, Croatia, October 2019
- › *Dynamic consensus community detection and combinatorial multi-armed bandit*, IEEE/ACM Int. Conference on Advances in Social Networks Analysis and Mining (**ASONAM '19**), Vancouver, Canada, August 2019

Tutorials

- › Domenico Mandaglio, Lucio La Cava and Andrea Tagarelli. *Fair Clustering and Polarization Detection on Online Social Networks through Correlation Clustering*, 16th ACM Web Science Conference (**WebSci '24**), Stuttgart, Germany, May 2024

Workshops

- › *Polarized Communities meet Densest Subgraph: Efficient and Effective Polarization Detection in Signed Networks*, International Workshop on Sustainable Artificial Intelligence for addressing Online Information Disorder (**SAI4OID '26**), Naples, Italy, September 2026
- › *Polarized Communities in Mastodon: Insights from Instance-Level Analysis*, International Workshop on Sustainable Artificial Intelligence for addressing Online Information Disorder (**SAI4OID '25**), London, UK, November 2025
- › *Correlation Clustering with Fairness Constraints*, Workshop on Green-Aware Artificial Intelligence (**GreenAI '25**), Bologna, Italy, October 2025
- › *Theoretical basis and computational complexity of semifactual explanations*, National Conference on Artificial Intelligence (**Ital-IA '25**), Trieste, Italy, June 2025
- › *Graph Deep Learning for Green-aware Knowledge*, 1st Workshop of the Spoke 9 of the PNRR Future AI Research (FAIR) project, Rende, Italy, February 2024

EXTERNALLY-FUNDED RESEARCH PROJECTS

Participation as a research-staff member

- › *Future Artificial Intelligence Research (FAIR), Spoke 9 - Green-Aware AI*
Funding entity: MUR PNRR 2023-2025
CUP: H23C22000860006
- › *CATCH 4.0 – An intelligent Consumer-centric Approach To manage engagements, Contents & insigHts*
Funding entity: Ministry of Economic Development (MISE)
CUP: B78I20000130005
- › *D-All: sfruttare la potenza dei big data per effettuare analisi predittiva in applicazioni innovative di cross-market*
Funding entity: Italian PON 2018-2020
CUP: B48I17000820008

TEACHING ACTIVITY

University Courses (Instructor)

- › **Data Mining**
University of Calabria, Rende (Italy)
B.Sc. in Computer Science & Engineering
Academic years: '25-'26 (38 hours), '24-'25 (26 hours), '23-'24 (24 hours)
- › **Database Systems and Data Analytics**
University of Calabria, Rende (Italy)
B.Sc. in Statistics for Data Science
Academic years: '25-'26 (15 hours), '24-'25 (15 hours), '23-'24 (16 hours)
- › **Data Mining and Machine Learning on Graphs (Invited Lecturer)**
École Polytechnique Féminine (EPF) Engineering School, Montpellier (France)
M.Sc. in Data Science
Academic years: '25-'26 (24 hours)
- › **Informatics and Elements of Medical Informatics**
University of Calabria, Rende (Italy)
single-cycle M.Sc. in Medicine and Surgery Digital Technologies
Academic years: '23-'24 (33 hours)
- › **Correlation Clustering: from classic to reinforcement learning-based theoretical frameworks, with applications to fairness and diversity related problems**
University of Calabria, Rende (Italy)
PhD program in Information and Communication Technologies
Academic years: '22-'23 (10 hours)

University Courses (Teaching Assistant)

- › **Data Mining**
University of Calabria, Rende (Italy)
B.Sc. in Computer Science & Engineering
Academic years: '25-'26 (18 hours)
- › **Big Data Analytics**
University of Calabria, Rende (Italy)
M.Sc. in Data Science for Business Analytics
Academic years: '25-'26 (14 hours), '24-'25 (14 hours)
- › **Database Systems and Data Analytics**
University of Calabria, Rende (Italy)
B.Sc. in Statistics for Data Science
Academic years: '25-'26 (6 hours), '24-'25 (6 hours), '23-'24 (5 hours)
- › **Algorithms and Data Structures (Group A)**
University of Calabria, Rende (Italy)
B.Sc. in Computer Science & Engineering
Academic years: '22-'23 (18 hours), '21-'22 (18 hours)
- › **Algorithms and Data Structures (Group B)**
University of Calabria, Rende (Italy)
B.Sc. in Computer Science & Engineering

Academic years: '21-'22 (18 hours)

› **Algorithms and Data Structures**

University of Calabria, Rende (Italy)

B.Sc. in Computer Science & Engineering

Academic years: '20-'21 (18 hours), '18-'19 (18 hours), '17-'18 (18 hours)

› **Database Systems**

University of Calabria, Rende (Italy)

M.Sc. in Management Engineering

Academic years: '19-'20 (15 hours), '18-'19 (15 hours)

Other Teaching Activities

› **Cybersecurity and the Internet of Things: Security in a Cyber-Physical and Interconnected World**

University of Calabria, Rende (Italy)

Seminar for high-school students

Academic years: '25-'26 (1 hour), '24-'25 (1 hour)

› **Databases, Information Systems, and Processes for Public Administration**

National School of Administration (SNA), Calabria Territorial Training Centre, Reggio Calabria (Italy)

Lecture within the course *Digital Transformation: Process Management and Artificial Intelligence (AI) Support*

Academic years: '25-'26 (3 hours)

B.Sc./M.Sc. Thesis Advisorships

- › *Design and Development of an LLM- and RAG-Based Chatbot to Support Web InBasket Simulations*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '25-'26)
- › *Fairness in Language Models: Bias, Evaluation Methods, and Mitigation Strategies*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Design and Development of a Cloud-Native Platform for OpenCoesione Data Processing*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Machine Learning for Optimizing Debt Collection Processes*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Locality-Sensitive Hashing for k-Nearest Neighbors Problems*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Design and Development of ArkHub: A Collaborative Platform for Reusing VHDL Components*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Machine Learning and Formula 1: Predictive Models for Race Results*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Tourism and Artificial Intelligence: An LLM- and RAG-Based Chatbot for Personalized Travel Assistance*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Design and Development of an AI-Based Web Platform for Application Management and Personnel Selection*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '24-'25)
- › *Advances in Asset Pricing: Leveraging Machine Learning for Asset Valuation*,
University of Calabria, B.Sc. in Statistics for Data Science (A.Y. '24-'25)
- › *Study and Analysis of Graph Neural Network Techniques for Signed Graphs*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '23-'24)
- › *Study and Analysis of Active Learning Techniques*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '23-'24)
- › *Methods for Identifying Polarized Communities in Signed Networks*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '23-'24)
- › *A Spectral Algorithm for Extracting Dense Subgraphs*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '23-'24)
- › *Discovering Polarized Communities in Signed Graphs*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '23-'24)
- › *Clustering for Malware Detection*,
University of Calabria, B.Sc. in Computer Science & Engineering (A.Y. '23-'24)

HONORS & AWARDS

- › **NeurIPS 2025 Top Reviewer**: granted to the top 8% of reviewers

- > **WebSci '24 Best Short Paper Award** for the paper “Polarization in Decentralized Online Social Networks” [14]
- > **Fair Access Fund: Travel Grant**, The ACM Web Science Conference 2024
- > **WWW 2023 Best Reviewer Award**: granted to the top 10% of reviewers “selected by track chairs of the research track from the Program Committee and Senior Program Committee, based on the quality of their reviews and metareviews, and their engaged participation during the discussion phase”

SKILLS (NON-EXHAUSTIVE)

General-Purpose Programming Languages: Python, Java, C

Domain-Specific Languages: SQL (relational DB querying), Bash (shell scripting), LaTeX (typesetting)

Versioning Technology: Git

Interacting Computing Technology: Jupyter Notebook

Software Libraries: numpy, scikit-learn, PyTorch, PyTorch Geometric, Hugging Face Transformers, pandas, seaborn, NetworkX, igraph

Operating Systems: macOS, Windows, Unix/Linux

Language Skills: (Native) Italian, (Fluent) English, (Basic) French

PUBLICATIONS

Journals

- [1] Francesco Gullo, Domenico Mandaglio, and Andrea Tagarelli. “Top-k Diverse Polarized Communities in Signed Networks”. *Machine Learning (MACH)* 115.7 (2026), p. 166. DOI: 10.1007/S10994-026-07098-8.
- [2] Francesco Gullo, Domenico Mandaglio, and Andrea Tagarelli. “Polarized Communities Meet Densest Subgraph: Efficient and Effective Polarization Detection in Signed Networks”. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 20.2 (2026), pp. 1–26. DOI: 10.1145/3779064.
- [3] Lucio La Cava, Domenico Mandaglio, Lorenzo Zangari, and Andrea Tagarelli. “Heuristic-informed mixture of experts for link prediction in multilayer networks”. *Information Sciences (Inf. Sci.)* 736 (2026), p. 123106. DOI: 10.1016/J.INS.2026.123106.
- [4] Sergio Flesca, Domenico Mandaglio, and Francesco Scala. “Enhancing active learning through latent space exploration: A k-nearest neighbors approach”. *Array* 28 (2025), p. 100584. DOI: 10.1016/J.ARRAY.2025.100584.
- [5] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and Irina Trubitsyna. “Decentralized federated learning meets Physics-Informed Neural Networks”. *Knowledge-Based Systems (KBS)* 323 (2025), p. 113717. DOI: 10.1016/j.knosys.2025.113717.
- [6] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, and Irina Trubitsyna. “Abstract argumentation frameworks with strong and weak constraints”. *Artificial Intelligence (AIJ)* 336 (2024), p. 104205. DOI: 10.1016/j.artint.2024.104205.
- [7] Francesco Gullo, Domenico Mandaglio, and Andrea Tagarelli. “Neural Discovery of Balance-aware Polarized Communities”. *Machine Learning (MACH)* 113.9 (2024), pp. 6611–6644. DOI: 10.1007/s10994-024-06581-4.
- [8] Sergio Flesca, Domenico Mandaglio, Francesco Scala, and Andrea Tagarelli. “A meta-active learning approach exploiting instance importance”. *Expert Systems with Applications (ESWA)* 247 (2024), p. 123320. DOI: 10.1016/j.eswa.2024.123320.
- [9] Francesco Gullo, Domenico Mandaglio, and Andrea Tagarelli. “A Combinatorial Multi-Armed Bandit Approach to Correlation Clustering”. *Data Mining and Knowledge Discovery (DAMI)* 37.4 (2023), pp. 1630–1691. DOI: 10.1007/s10618-023-00937-5.
- [10] Domenico Mandaglio and Andrea Tagarelli. “Generalized Preference Learning for Trust Network Inference”. *IEEE Access* 7 (2019), pp. 174583–174596. DOI: 10.1109/ACCESS.2019.2957191.

International Conferences

- [11] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and Irina Trubitsyna. “Semifactual Explanations for GNN-based Classification: Formal Foundations, Complexity and Computation”. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI-ECAI '26)*, Bremen, Germany, August 15-22, 2026.
- [12] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, and Irina Trubitsyna. “Conditional Probabilistic Bipolar Argumentation Framework: Explanations, Complexity and Approximation”. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI '26)*, Singapore, January 20-27, 2026, pp. 18910–18919.

- [13] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and Irina Trubitsyna. “Even-if Explanations: Formal Foundations, Priorities and Complexity”. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI ’25)*, Philadelphia, Pennsylvania, USA, February 25 - March 4, 2025, pp. 15347–15355.
- [14] Lucio La Cava, Domenico Mandaglio, and Andrea Tagarelli. “Polarization in Decentralized Online Social Networks”. In *Proceedings of the ACM Web Science Conference (WebSci ’24)*, Stuttgart, Germany, May 21-24, 2024, pp. 48–52.
- [15] Lorenzo Zangari, Domenico Mandaglio, and Andrea Tagarelli. “Link Prediction on Multilayer Networks through Learning of Within-Layer and Across-Layer Node-Pair Structural Features and Node Embedding Similarity”. In *Proceedings of the ACM Web Conference (WWW ’24)*, Singapore, May 13-17, 2024, pp. 924–935.
- [16] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, and Irina Trubitsyna. “Complexity of verification and existence problems in epistemic argumentation framework”. In *Proceedings of the European Conference on Artificial Intelligence (ECAI ’23)*, Krakow, Poland, September 30 - October 4, 2023, pp. 77–84.
- [17] Francesco Gullo, Lucio La Cava, Domenico Mandaglio, and Andrea Tagarelli. “When Correlation Clustering Meets Fairness Constraints”. In *Proceedings of the International Conference on Discovery Science (DS ’22)*, Montpellier, France, October 10-12, 2022, pp. 302–317.
- [18] Sergio Flesca, Domenico Mandaglio, Francesco Scala, and Andrea Tagarelli. “Learning to Active Learn by Gradient Variation Based on Instance Importance”. In *Proceedings of the International Conference on Pattern Recognition (ICPR ’22)*, Montréal, Québec, Canada, August 21-25, 2022, pp. 2224–2230.
- [19] Domenico Mandaglio, Andrea Tagarelli, and Francesco Gullo. “Correlation Clustering with Global Weight Bounds”. In *Proceedings of the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD ’21)*, Bilbao, Spain, September 13-17, vol. 12976. 2021, pp. 499–515.
- [20] Domenico Mandaglio, Andrea Tagarelli, and Francesco Gullo. “In and Out: Optimizing Overall Interaction in Probabilistic Graphs under Clustering Constraints”. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD ’20)*, Virtual Event, CA, USA, August 23-27, 2020, pp. 1371–1381.
- [21] Domenico Mandaglio and Andrea Tagarelli. “Dynamic consensus community detection and combinatorial multi-armed bandit”. In *Proceedings of the IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM ’19)*, Vancouver, British Columbia, Canada, 27-30 August, 2019, pp. 184–187.
- [22] Domenico Mandaglio and Andrea Tagarelli. “A Combinatorial Multi-Armed Bandit Based Method for Dynamic Consensus Community Detection in Temporal Networks”. In *Proceedings of the International Conference on Discovery Science (DS ’19)*, Split, Croatia, October 28-30, 2019, pp. 412–427.
- [23] Domenico Mandaglio, Alessia Amelio, and Andrea Tagarelli. “Consensus Community Detection in Multilayer Networks Using Parameter-Free Graph Pruning”. In *Proceedings of the Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD ’18)*, Melbourne, VIC, Australia, June 3-6, 2018, pp. 193–205.

National Conferences & Workshops

- [24] Lucio La Cava, Domenico Mandaglio, and Andrea Tagarelli. “Polarized Communities in Mastodon: Insights from Instance-Level Analysis”. In *Proceedings of the 1st International Workshop on Sustainable Artificial Intelligence for addressing Online Information Disorder (SAI4OID ’25) co-located with the 24th International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT ’25)*, London, UK, November 15-18, 2025.
- [25] Francesco Gullo, Lucio La Cava, Domenico Mandaglio, and Andrea Tagarelli. “Correlation Clustering with Fairness Constraints”. In *Proceedings of the 2nd Workshop on Green-Aware Artificial Intelligence (GreenAI ’25) co-located with the 28th European Conference on Artificial Intelligence (ECAI ’25)*, Bologna, Italy, October 25-30, 2025.
- [26] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and Irina Trubitsyna. “Theoretical basis and computational complexity of semifactual explanations”. In *Joint Proceedings of the Thematic Workshops co-located with the 5th National Conference on Artificial Intelligence (Ital-IA ’25)*, Trieste, Italy, June 23-24, 2025.
- [27] Francesco Gullo, Domenico Mandaglio, and Andrea Tagarelli. “Discovering Balance-Aware Polarized Communities in Signed Networks with Graph Neural Networks”. In *Proceedings of the 33rd Italian Symposium on Advanced Database Systems (SEBD ’25)*, Ischia, Italy, June 16-19, 2025.
- [28] Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, and Irina Trubitsyna. “On the Verification and Existence Problems in Epistemic Argumentation Framework”. In *Proceedings of the 7th Workshop on Advances in Argumentation in Artificial Intelligence (AI³ ’23) co-located with the 22nd International Conference of the Italian Association for Artificial Intelligence (AIIA ’23)*, Rome, Italy, November 9, vol. 3546. 2023.

- [29] Sergio Flesca, Domenico Mandaglio, Francesco Scala, and Andrea Tagarelli. “A Meta-Active Learning Approach Exploiting Instance Importance based on Learning Gradient Variation”. In *Proceedings of the 31th Italian Symposium on Advanced Database Systems (SEBD '23)*, Galzignano Terme, Italy, July 2-5, 2023, pp. 535–544.
- [30] Domenico Mandaglio, Andrea Tagarelli, and Francesco Gullo. “Correlation Clustering: from Local to Global Constraints”. In *Proceedings of the 30th Italian Symposium on Advanced Database Systems (SEBD '22)*, Tirrenia (Pisa), Italy, June 19-22, 2022, pp. 445–452.

Book Chapters

- [31] Domenico Mandaglio and Andrea Tagarelli. “Data Mining: Clustering”. In *Encyclopedia of Bioinformatics and Computational Biology, 2nd Edition*. Elsevier, 2025. ISBN: 978-0-12-809633-8. DOI: 10.1016/B978-0-323-95502-7.00171-8.